

Roshan Hegde

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EDUCATION

University of Illinois at Urbana-Champaign

May 2026

B.S. in Computer Science, Grainger College of Engineering

GPA: 4.0/4.0

B.S. in Mathematics, College of Liberal Arts & Sciences

Highland Park High School

May 2023

GPA: 4.69/4.0

RELEVANT COURSEWORK

Courses: Algorithms, Operating Systems Design, Programming Languages and Compilers, Algorithms and Models of Computation, Cryptography, Computer Architecture, Numerical Methods, Data Structures (Honors), Probability and Statistics for Computer Science, Fundamental Mathematics, Abstract Linear Algebra, Linear Algebra with Computational Applications,

Northwestern University Courses: Java, C++, Machine Learning

SKILLS

Languages: C, C++, Go, Java, Kotlin, Python, Lua, Rust, Javascript, Typescript, HTML, CSS

Tools: Git, GitHub, Docker, Linux, Nix, Vim

Frameworks: Svelte, React, Tailwindcss, Vite, AWS

Libraries: numpy, pytorch, pandas, matplotlib

AWARDS

1st place, MechMania '24 | *Java, Python, Git*

September 2024

- Competed in a 24-hour Game AI Hackathon held annually since 1995 where students' AI models compete

1st place, Hackillinois '24 | *Rust, Cargo, Nix, Git*

February 2024

- Competed against over 1,100 other students at Illinois' largest Hackathon
- Built a developer tool used to universally initialize a basic development environment for over 30 languages
- Designed modular software architecture allowing for native configuration and rapid extensibility

1st place, MechMania '23 | *Java, Python, Git*

September 2023

- Implemented A* pathfinding with a modified Manhattan distance heuristic
- Employed a divide and conquer strategy in tandem with pathfinding to outperform other models

PROJECTS

nix-tinker | *Rust, Cargo, Nix, Nix Flakes*

May 2024 – Present

- Developed an open-source tool for rapid prototyping of Nix configuration files, streamlining development workflow
- Crafted an intuitive CLI for managing file links with the Nix store, significantly enhancing user efficiency
- Leveraged Nix Flakes to guarantee reproducible and consistent build environments across diverse systems

RecursiveTicTacToeML | *Rust, Cargo, Nix, Git*

June 2022 – Present

- Designed and implemented an n-dimensional recursive Tic Tac Toe game to explore advanced game dynamics
- Developed a self-training AI using custom gradient descent and backpropagation, generating data through self-play
- Engineered a real-time, websocket-based multiplayer mode with rigorous game state validation

EXPERIENCE

MechMania | *Lead Infrastructure Developer* | *Go, AWS, python*

August 2024 – Present

- Built a multithreaded web-socket server in Go interacting with AWS services including databases and cloud storage
- Implemented a Java-Python language interop system for broader audience reach

Opensource@Illinois | *Member*

August 2024 – Present

- Attended workshops focused on teaching good open source development practices and tools
- Working in a subteam to contribute to various open source projects